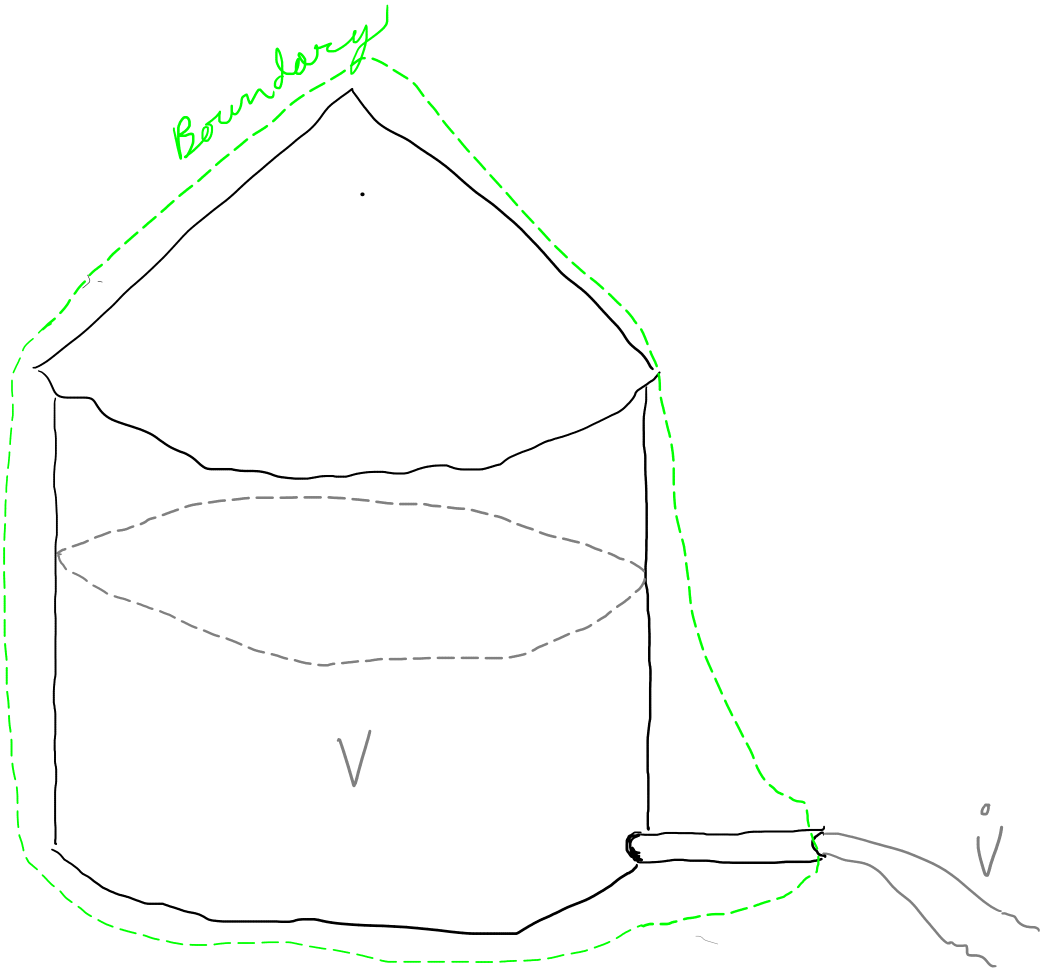
Water tank exercise solution

Modelling 2025

1.



2.

$$\text{accumulation} = \cancelto{0}{\text{flow in}} - \text{flow out} + \cancelto{0}{\text{generation}}$$

$(m^3\ s^{-1})$

$$\text{accumulation} = - \text{flow out}$$

3.

volume of water remaining

$$\underbrace{\text{Rate of flow out}}_{\dot{V}} = c \cdot V$$

$(m^3\ s^{-1})$

c – a proportionality constant

4. No linkage equations needed.

5. We don't need to think about space. Our constitutive equation applies to one location - the exit pipe. Our mass (volume) applies to the entire tank.

6. $V_0 = 10\ m^3$ But we don't need this unless we want to make predictions.

7. Governing equation for this exercise should look like

$$\frac{dV}{dt} = f(V)$$

Just put equations from 2 & 3 together!

accumulation $\rightarrow$ $\frac{dV}{dt}$

$$\frac{dV}{dt} = -cV$$

8. Integrate (and check if unsure)

$$\boxed{V(t) = V_0 e^{-ct}}$$

Done! Can you implement it in Python?

Going a bit further...

We know V_0 = initial volume = $10\ m^3$.

How about the rate constant c ?

We also know $\left.\frac{dV}{dt}\right|_{t=0} = -0.1\ m^3\ min^{-1} = f_0$

So $\left.\frac{dV}{dt}\right|_{t=0} = -cV_0 = \dot{V}_0$

$$c = \frac{\dot{V}_0}{V_0} = \frac{0.1\ m^{3}\ min^{-1}}{10\ m^3} = 0.01\ min^{-1}$$

So now we have both constants.

Could we make the solution more general?

We can recognize that

$\dot{V} = c' \frac{V}{A}$ because pressure drives flow

and pressure is proportional to water height above the exit (hydraulic head).

So $V(t) = V_0 e^{-\frac{c'}{A}t}$ or $\boxed{\frac{V}{V_0} = e^{-\frac{c'}{A}t}}$

This model solution could be applied to <u>any</u> tank with vertical sides <u>and</u> an outlet with a known c'.

Do its predictions make sense?

Time to e.g. 50% loss depends <u>only</u> on c' and A.